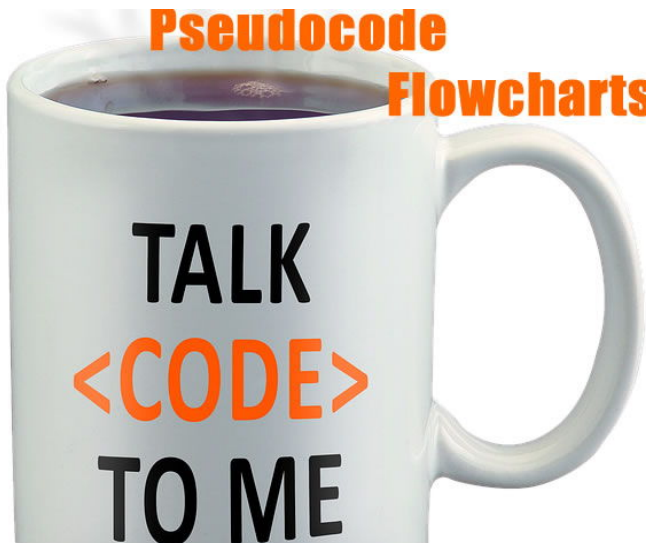


Computer Science

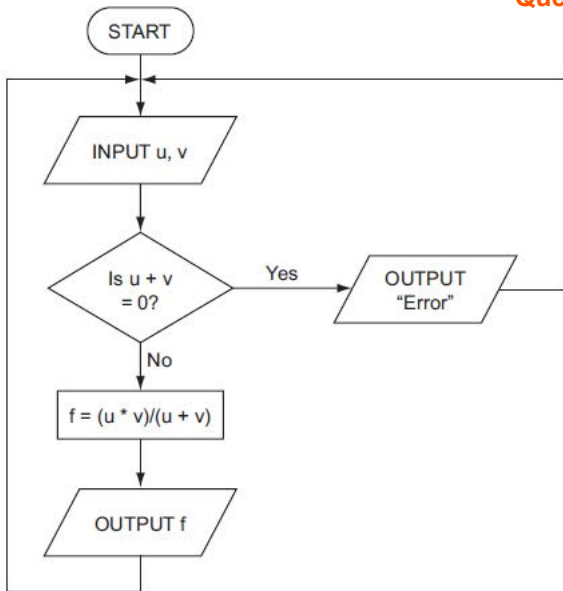
Pseudocode

Flowcharts



The following flowchart inputs **two** numbers, carries out a calculation and then outputs the result.

Question - 1



(a) Complete the following table for the **three** sets of input data.

INPUT		OUTPUT
u	v	
5	5	
6	-6	
12	4	

[3]

(b) The above algorithm has been placed in a library of routines. Give **one** advantage of doing this.

.....

.....

.....

[1]

Algorithms and programs use loops to control the number of times a particular procedure is used.

Question - 2

Two methods are **repeat ... until** and **for ... to**.

(a) Write a procedure using both these loop methods to input 20 numbers into a variable called **x**.

(i) **repeat ... until**

```
.....
.....
.....
.....
.....
```

[2]

(ii) **for ... to**

```
.....
.....
.....
.....
.....
```

[2]

(b) Name another loop structure.

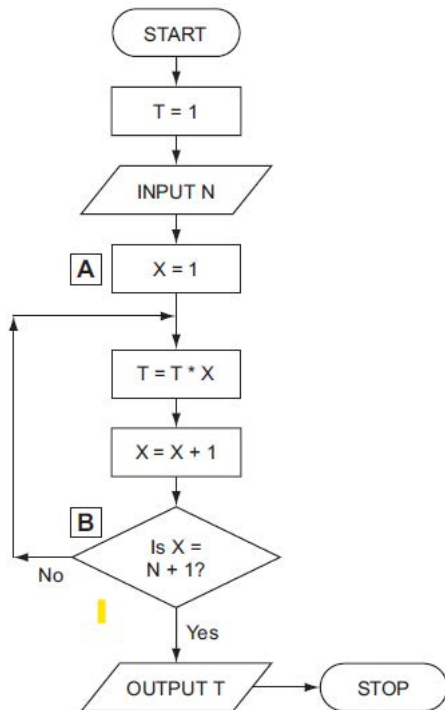
```
.....
.....
```

[1]

Question - 3

- withdrawal is refused if amount entered > current balance
- withdrawal is refused if amount entered > daily limit
- if current balance < \$100, then a charge of 2% is made
- if current balance $\geq$ \$100, no charge is made

[5]



(a) Complete the table to show what outputs you would expect for the **two** inputs.

Input N	Output T
5	
1	

[2]

(b) Write down a possible LOOP construct for the section A to B in the flowchart using pseudocode.

.....

.....

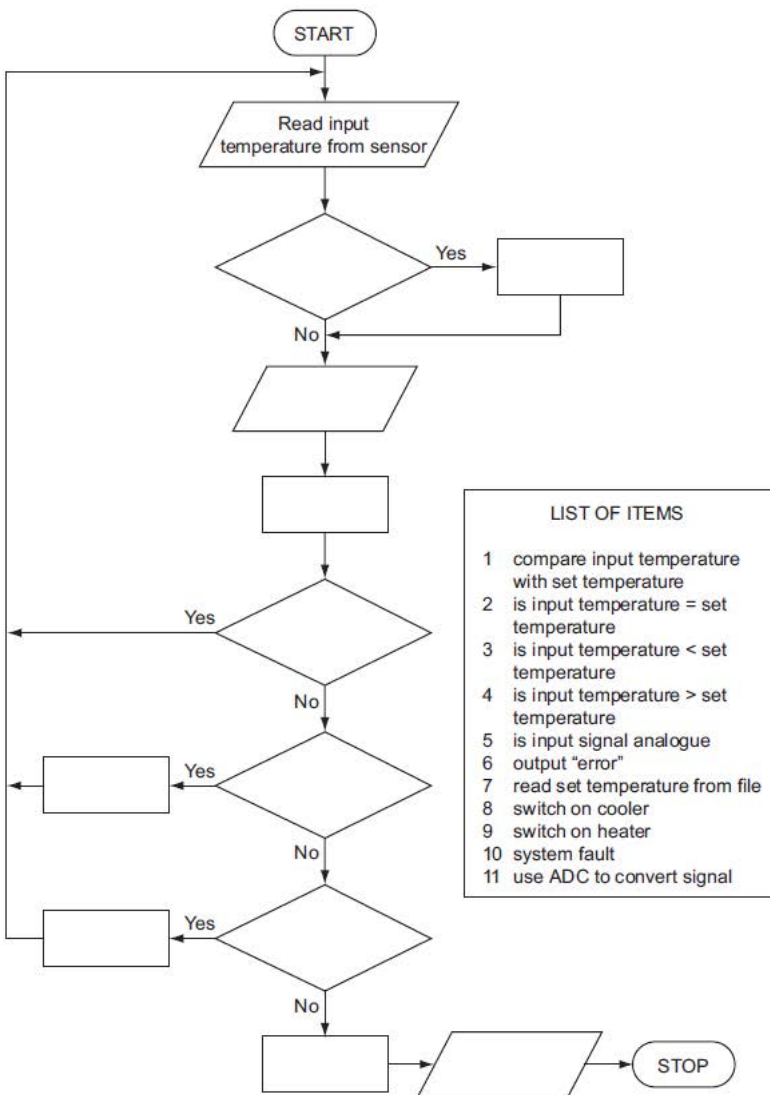
.....

.....

[2]

The following flowchart shows how sensors (which can be analogue or digital) and a computer are used to control the temperature of a greenhouse for plants. Complete the flowchart using the items from the list below.

Question - 6



The manufacturing cost of producing an item depends on its complexity. A company manufactures three different types of item, with costs based on the following calculations:

Item type 1: item cost = parts cost * 1.5
Item type 2: item cost = parts cost * 2.5
Item type 3: item cost = parts cost * 5.0

Question - 7

The company makes 1000 items per day.

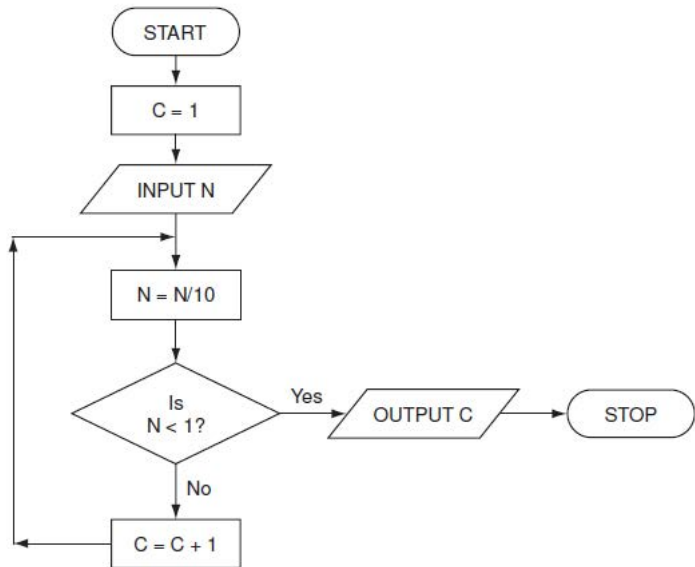
Write an algorithm, using pseudocode, flowchart or otherwise, which

- inputs the item type and parts cost of each item
- outputs the item cost for each item
- calculates and outputs the average (mean) item cost per day (based on 1000 items being made).

[5]

Study the flowchart.

Question - 8



Complete the table to show what outputs you would expect for the **three** inputs.

INPUT N	OUTPUT C
55	
2100	
1	

Write an algorithm, using pseudocode or otherwise, which inputs the times for 500 cars, calculates the final speed of each car using the formula in part (a), and then outputs:

- the final speed for ALL 500 cars
- the slowest (lowest) final speed
- the fastest (highest) final speed
- the average final speed for all the cars.

Question - 9

A group of students were monitoring the temperature every day over a one-year period. Readings were taken ten times every day (you may assume a year contains 365 days).

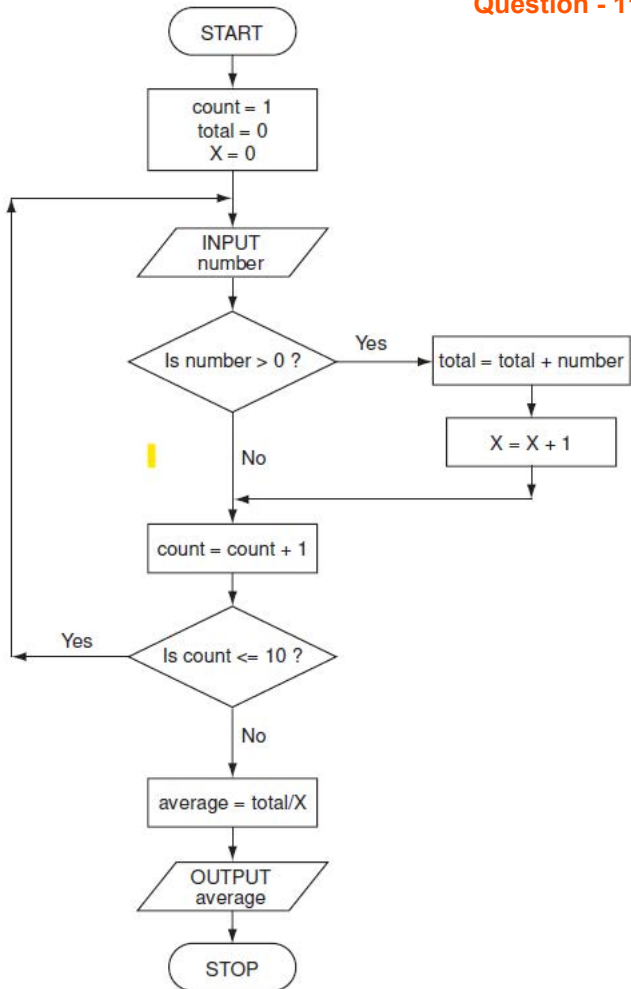
Write an algorithm, using pseudocode or flowchart, which

Question - 10

- inputs all the temperatures (ten per day)
- outputs the **highest** temperature taken over the year
- outputs the **lowest** temperature taken over the year
- outputs the average temperature **per day**
- outputs the average temperature **for the whole year**

Study the following flowchart very carefully:

Question - 11



(a) Complete the trace table for the following data set:

15, -2, 0, 8, 0, 21, -8, -12, 1, 25

count	number	total	X	average	OUTPUT

[4]

(b) What is the purpose of this flowchart?

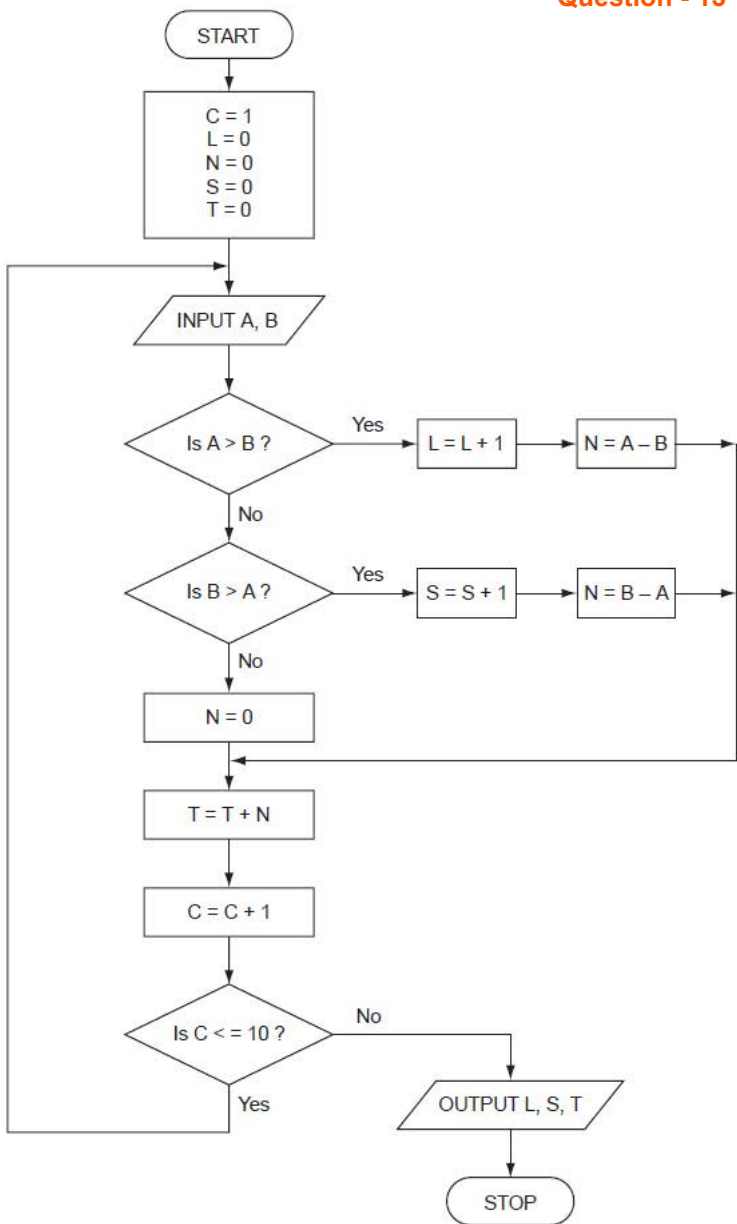
.....

.....

[1]

Carefully study the following flowchart.

Question - 13



(a) Complete the trace table for the following data:

8, 4, 3, 1, 5, 8, 4, 2, 1, 3, 2, 2, 1, 2, 5, 5, 4, 0, 5, 4

C	L	N	S	T	A	B

[6]

(b) What is the final output from the algorithm?

L =

S =

T =

[2]

- (a) Give **one** benefit and **one** drawback of using Internet banking.

Benefit

Drawback

[2]

She uses a 5-digit PIN.

- (b) Every time she logs on, she is asked to give 3 random digits from the PIN. She was asked to give her 3rd, 1st and 4th digit. This changes every time she logs on.

Give a reason for this.

[1]

- (c) A different application needs the whole PIN to be input.
The following code has been written to check the PIN:

```
c = 0
INPUT PIN
x = PIN
REPEAT
    x = x/10
    c = c + 1
UNTIL x < 1
IF c < 5
    THEN
        PRINT "error in PIN entered"
    ELSE
        PRINT "PIN OK"
ENDIF
```

- (i) What value of *c* and what message would be output if the following PINs were entered?

5 1 0 2 0 Value of *c*:

Message:

5 1 2 0 Value of *c*:

Message:

[2]

- (ii) What type of validation check is being carried out here?

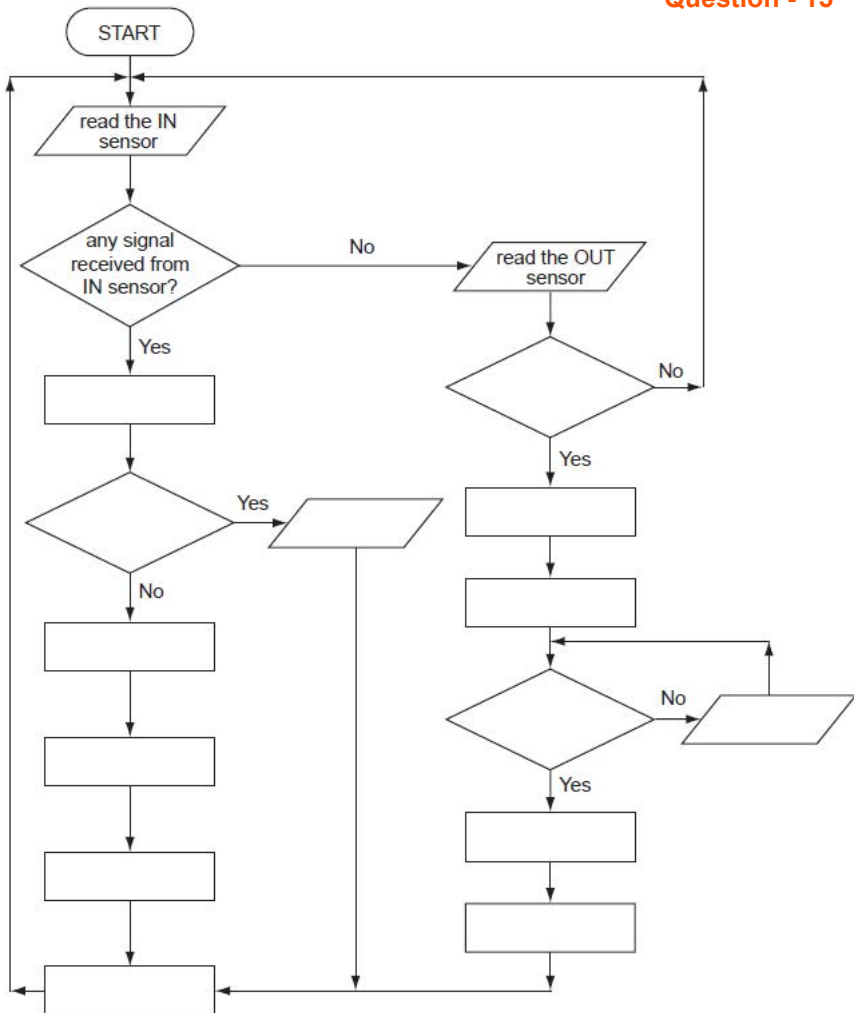
[1]

- 16 A car park uses sensors and a microprocessor to monitor cars leaving and entering. The car park is open 24 hours every day. The parking fee is \$10 per day.

The following flowchart shows how the IN and OUT barriers are controlled. Some of the statements are missing.

Using item numbers only, insert the correct item numbers into the flowchart from the item list.

Question - 15



List of statements

Item Number	Description
1	activate motor to raise IN barrier
2	activate motor to raise OUT barrier
3	any signal received from OUT sensor?
4	decrease number of cars in car park by 1
5	increase number of cars in car park by 1
6	is car park full?
7	is the car park fee paid?
8	OUTPUT "car park full"
9	OUTPUT "please pay car park fee at pay machine"
10	use ADC to convert IN sensor signal to digital
11	use ADC to convert OUT sensor signal to digital
12	use DAC to convert computer signal to analogue signal to operate IN barrier
13	use DAC to convert computer signal to analogue signal to operate OUT barrier
14	wait 30 seconds and then close barrier

A customer uses Internet banking. To gain access to their account they need:

- an 8-digit ID
- a 4-digit PIN
- a 10-character password

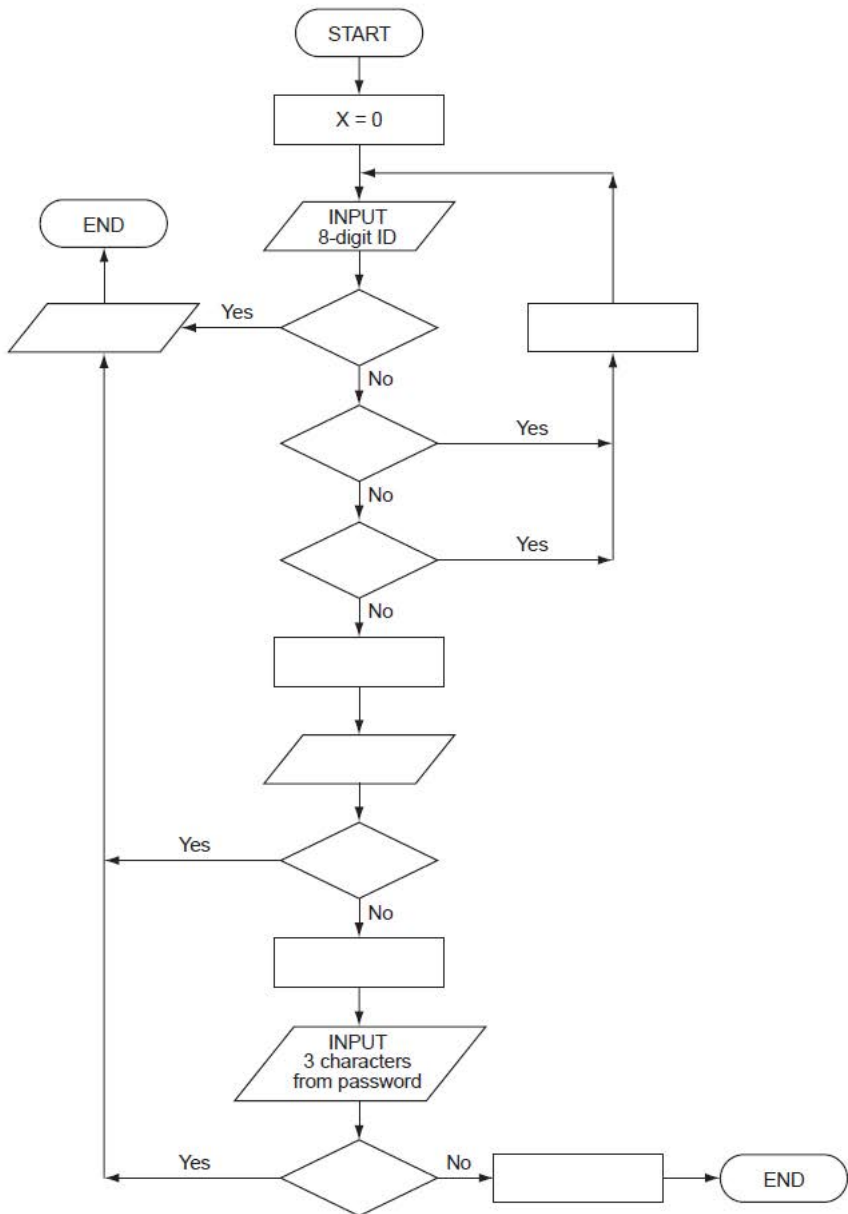
Question - 16

They will be asked to type in their ID, then 3 digits from their PIN and finally 3 characters from their password. Three attempts at the ID are allowed, but only one attempt at the PIN and at the password.

The flowchart on the next page shows the access process described above. However, most of the stages have been omitted.

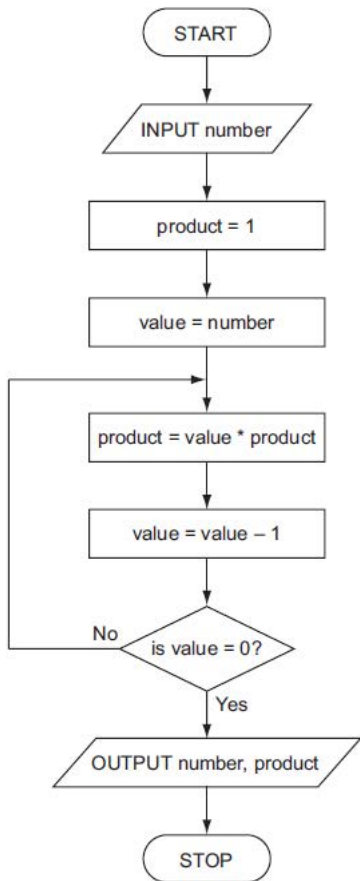
Complete the flowchart, **using item number only**, from the list of items given.

Item number	Item description
1	access to account allowed
2	are any characters in the password incorrect?
3	are any digits in the ID incorrect?
4	are any digits in the PIN incorrect?
5	generate three random digits from the PIN
6	generate three random characters from the password
7	input the required three digits from the PIN
8	is number of digits < 8 or number of digits > 8?
9	is $X > 2$?
10	output "access to account denied"
11	$X = X + 1$



Study the following flowchart very carefully.

Question - 17



Complete the trace table for the input value of 5:

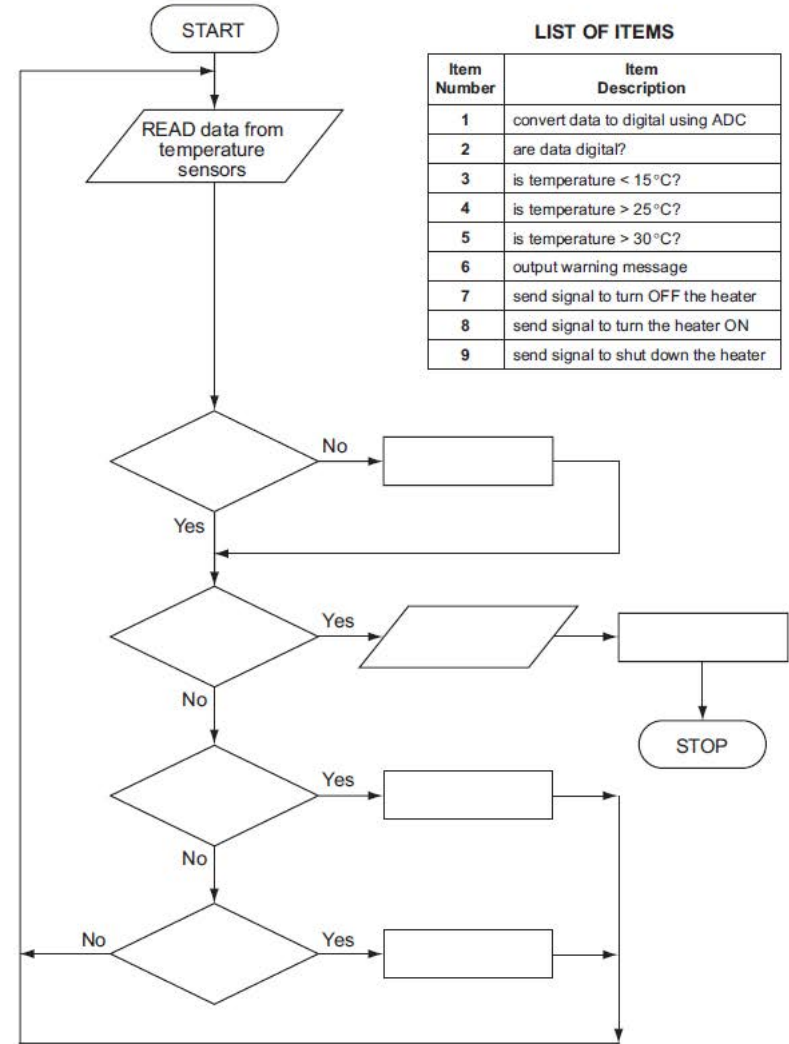
number	product	value	OUTPUT

A heating system is being controlled by sensors and a computer. The temperature must be kept between 15°C and 25°C. If 30°C is exceeded a warning message is generated and the system shuts down.

Question - 18

A flowchart of the process is shown below. Some of the items are missing.

Complete the flowchart, using item number only, from the list of items given.



A survey is being carried out which involves reading and recording sound levels near a busy road junction. Once all the data are collected, they are input manually into a computer. A sound level of 0 decibels (0 dB) is input to indicate the end of the data.

Question - 19

Write an algorithm, using pseudocode or a flowchart, which:

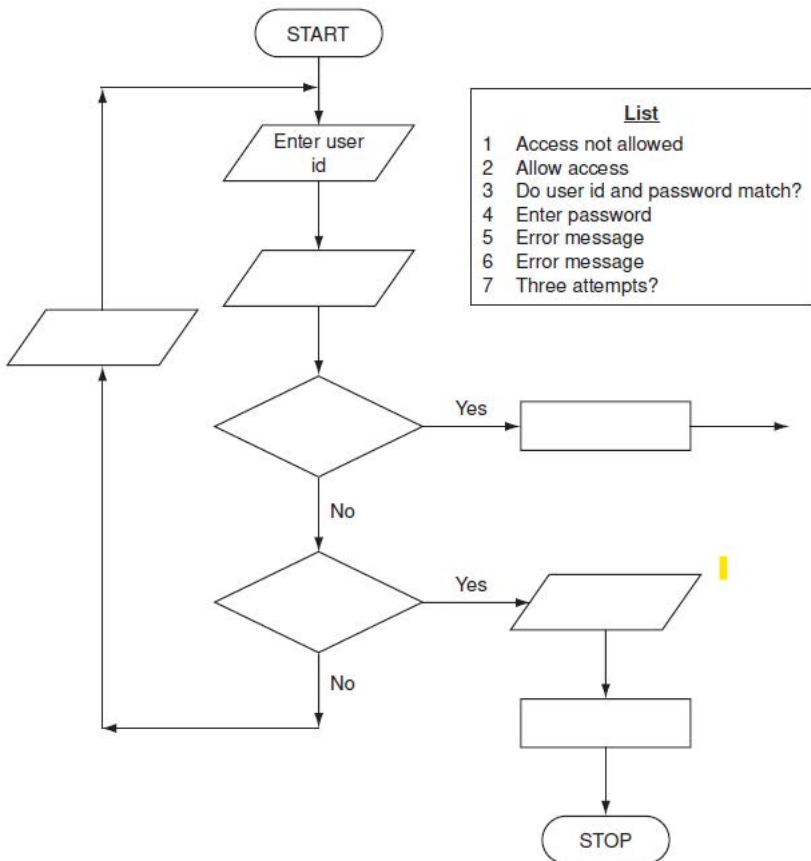
- inputs all the sound levels
- after a sound level of 0 is input, outputs the following:
 - average sound level
 - highest recorded sound level.

- 4 (a) To log on to a computer, a user needs to type in a user id followed by a password; these should match up. Only three attempts are allowed.

The flowchart below shows the log on procedure. Several boxes have been left blank.

Complete the flowchart using items from the list.

Question - 20



[3]

- (b) Sometimes it is necessary to key in the password twice.

Name this type of data check.

[1]

The following algorithm inputs 20 numbers and outputs how many numbers were positive (> 0) and how many numbers were negative (< 0).

Question - 21

```
1  negative = 1
2  positive = 1
3  for count = 1 to 20 do
4      input number
5      if number < 0 then negative = negative + 1
6      if number > 0 then positive = positive + 1
7      count = count + 1
8      print negative, positive
9  next count
```

There are **three** different errors in this algorithm.

Locate each error and give the reason why you think it is an error.

Error 1

Reason 1

Error 2

Reason 2

Error 3

Reason 3

Question - 22

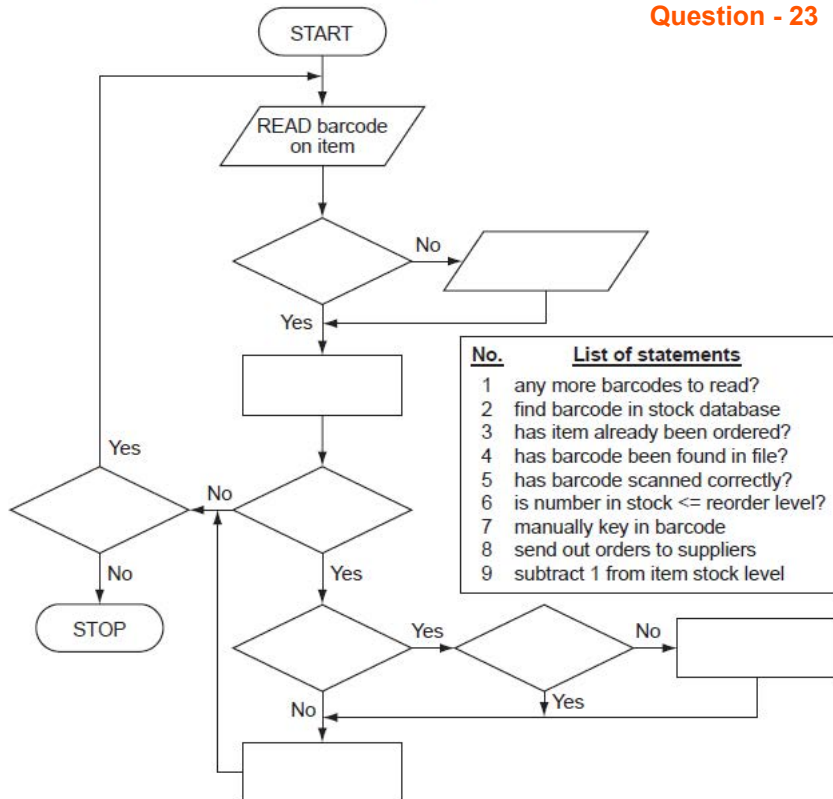
- inputs the height and weight of all 1000 students
- outputs the average (mean) height and weight
- includes any necessary error traps for the input of height and weight

[illegible]

- 4 The following flowchart shows how barcodes are used at the point of sale in an automatic stock control system.

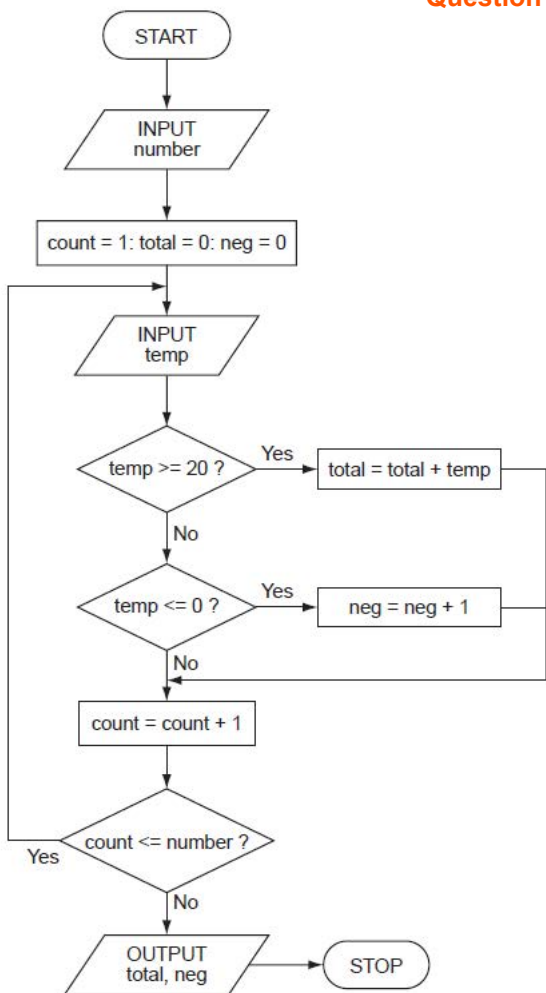
Select statements from the list below, using numbers only, to complete the flowchart.

Question - 23



Carefully study the following flowchart:

Question - 24



Complete the trace tables for the following two sets of test data:

(i) number = 7, temp = -5, 0, 5, -4, 0, 10, -2

(ii) number = 6, temp = 21, 20, 30, 19, 21, 15

(i) trace table:

number	count	temp	total	neg	OUTPUT

(ii) trace table:

number	count	temp	total	neg	OUTPUT

(a) Write an algorithm, using pseudocode or flowchart only, which:

- inputs three numbers
- outputs the **largest** of the three numbers

[3]

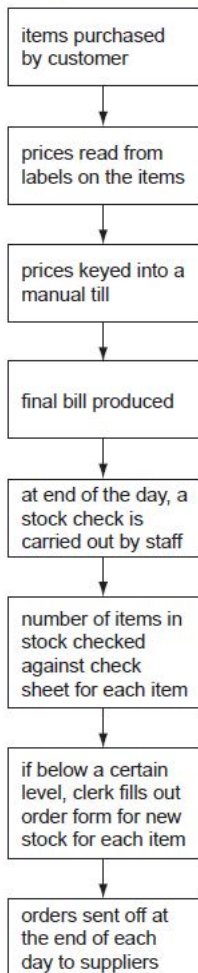
(b) Write an algorithm, using pseudocode or flowchart only, which:

- inputs 1000 numbers
- outputs how many of these numbers were whole numbers (integers)
(You may use $\text{INT}(X)$ in your answer e.g. $Y = \text{INT}(3.8)$ gives the value $Y = 3$)

[4]

Flowchart A

shows an existing manual system used by a shop to calculate a customer's bill and carry out stock control.

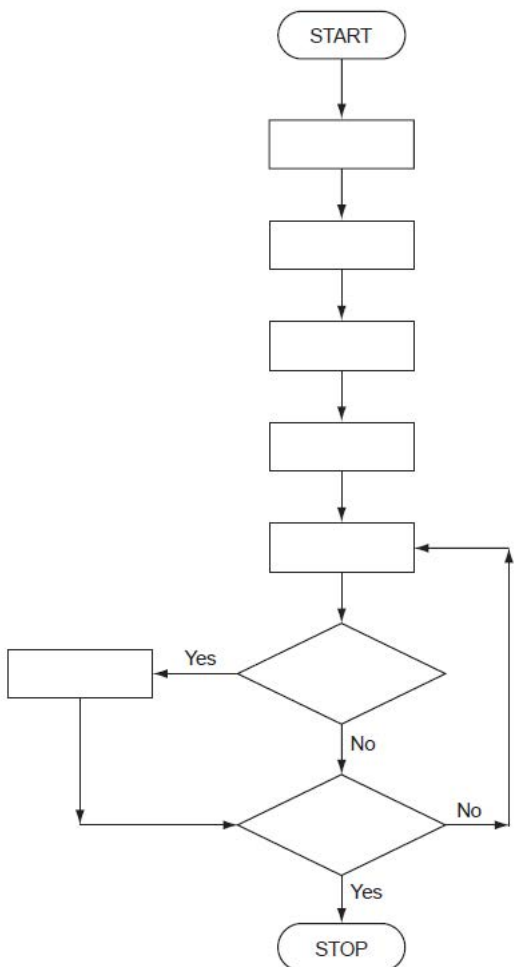


existing manual system

Question - 26

Flowchart B

shows the intended automatic computer-controlled system.



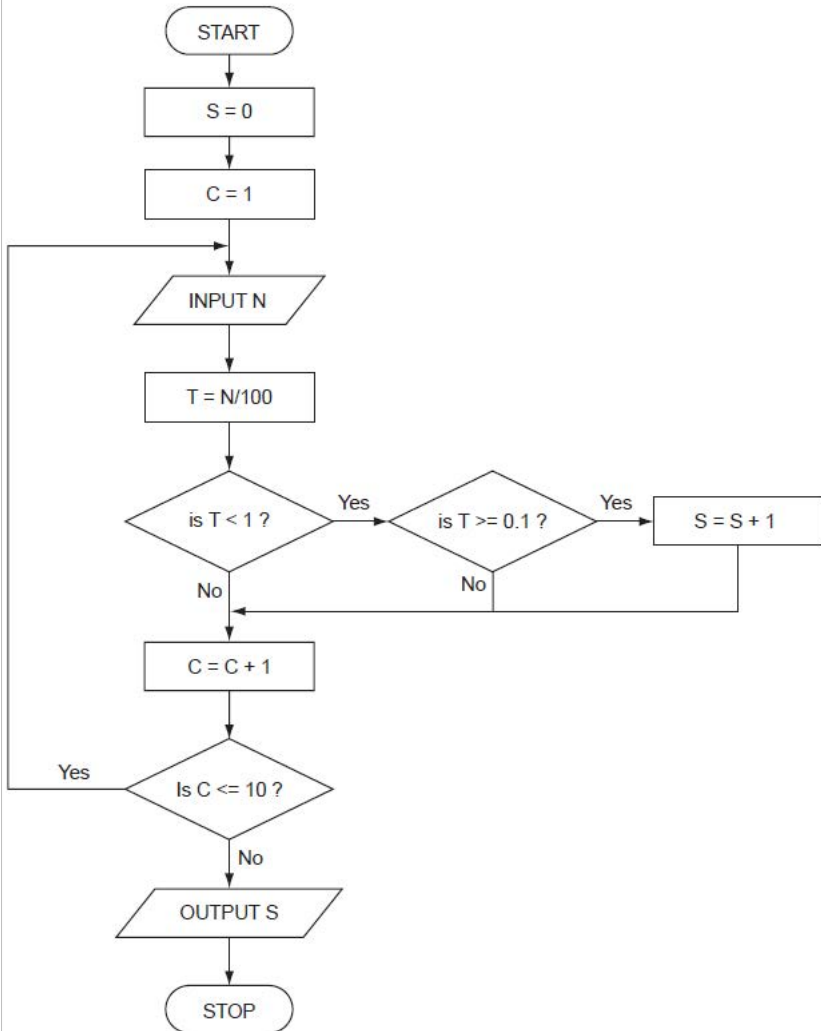
intended automatic computerised system

Using item numbers only from the following list, complete **Flowchart B**.

Item Number	Description
1	each item quantity reduced by 1 on the database
2	is stock level of item $\leq$ re-order level?
3	item price found on database
4	quantity of stock item checked against re-order level at end of the day
5	order for new stock of each item automatically sent out
6	barcode on each item scanned
7	have all items been checked?
8	an itemised bill is produced

Study the following flowchart very carefully.

Question - 27



Complete the trace table for the following data:

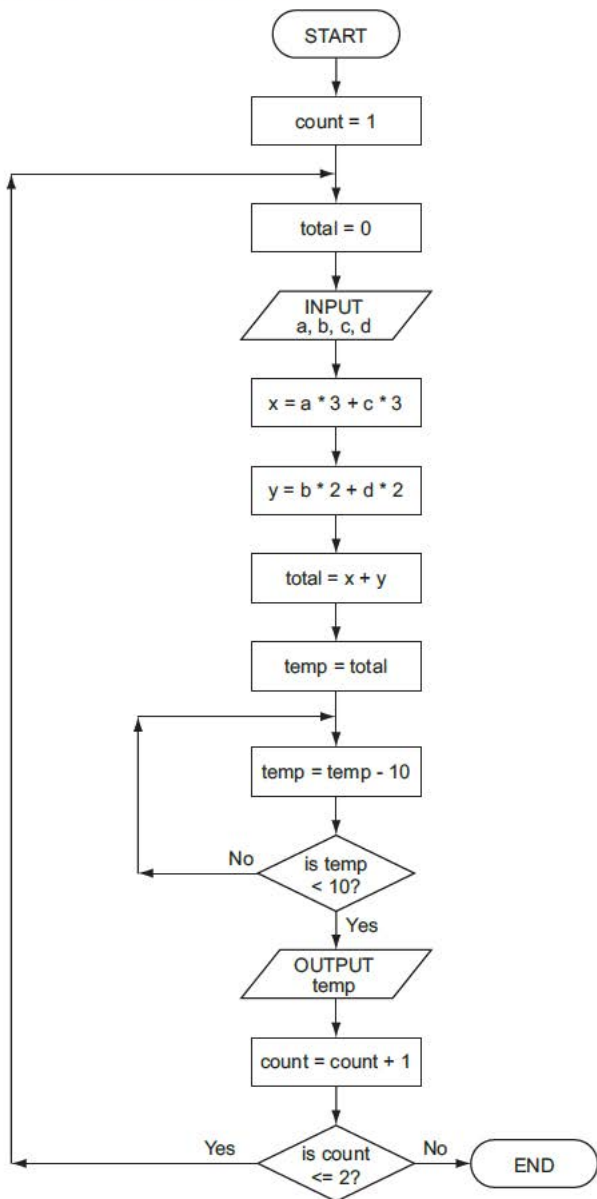
15, 8, 251, 35, 60, 3, 2, 1516, 19, 55

S	C	N	T	OUTPUT

(a) Write an algorithm, using pseudocode or a program flowchart only, that:

- inputs a series of positive numbers (-1 is used to terminate the input),
- outputs how many numbers were less than 1000 and
- outputs how many numbers were greater than 1000.

Question - 28



pacan

(i) $a = 5, b = 4, c = 1, d = 9$

(ii) $a = 5, b = 9, c = 4, d = 1$

[illegible]

A piece of pseudocode was written to input 1000 positive numbers and then output the highest and lowest numbers.

Question - 30

```
10 highest = 0
20 lowest = 0
30 for count = 1 to 100
40     input number
50     if number > highest then number = highest
60     if number < lowest then number = lowest
70     count = count + 1
80 next count
90 print highest, lowest
```

There are errors in the code.

Locate these errors and suggest a correction.

Error 1

Correction

Error 2

Correction

Error 3

Correction

Error 4

.....

.....

.....

Correction

.....

.....

.....

A library uses barcodes to identify borrowers and books. A borrower has a card containing a barcode with an equivalent numerical code. Each book also has a barcode with an equivalent numerical code.

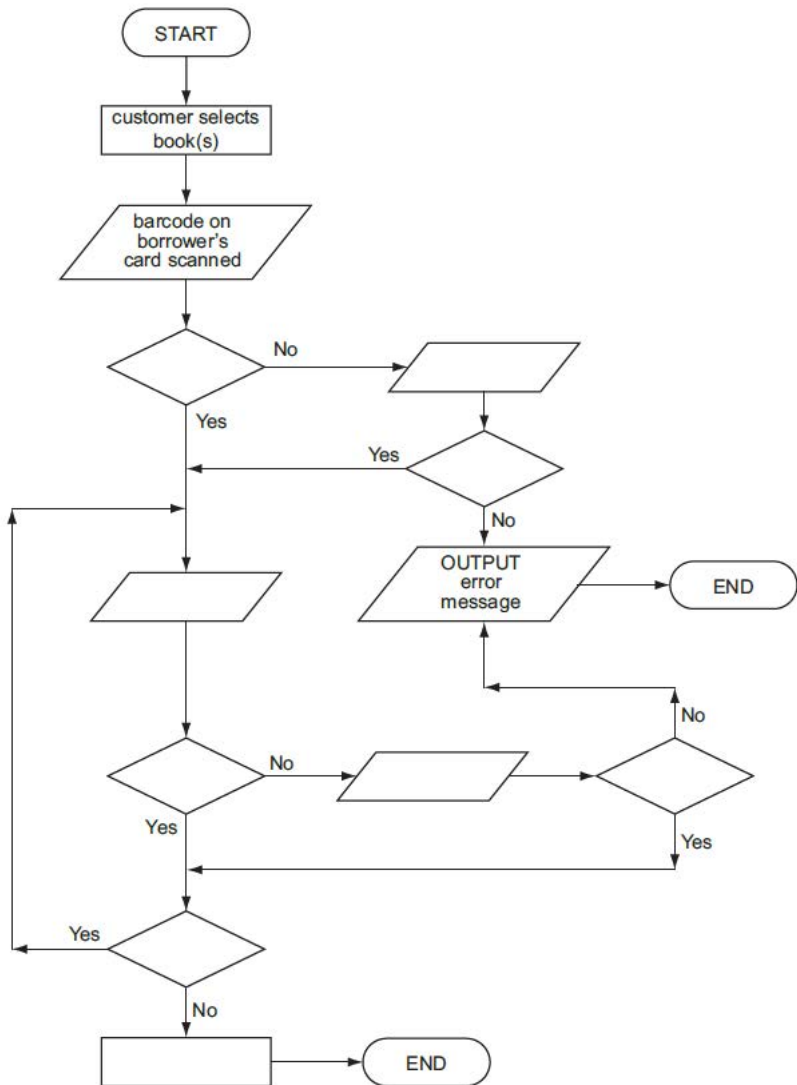
The flowchart on the facing page shows what happens when book(s) are taken out by a borrower. However, several stages are missing.

Using **item numbers only**, complete the flowchart by selecting stages from the item list below.

Question - 31

Item List

Item Number	Item Description
1	any more books to scan?
2	has book's barcode been recognised?
3	has book's numeric code been recognised?
4	has borrower's barcode been recognised?
5	has borrower's numeric code been recognised?
6	input book's numeric code manually
7	input borrower's numeric code manually
8	scan in barcode shown in book
9	update borrower and book files



The following section of a pseudocode algorithm should:

Question - 32

- input 500 numbers
- generate a ratio called **k**
- output each value of **k**
- output how many numbers were larger than 10

```
10 total = 1
20 FOR x = 1 TO 500
30     IF number < 10 THEN total = total + 1
40     k = x / number
50     x = x + 1
60     OUTPUT k
70 NEXT x
80 OUTPUT x
```

(a) There are five errors in the above code.

Locate these errors and suggest a correction.

error 1

correction

error 2

correction

error 3

correction

error 4

correction

error 5

correction

(b) The corrected algorithm was converted to a computer program and run. However, after several numbers were input, the program stopped and an error message was generated, showing that there was a further error at line 40 ($k = x / \text{number}$).

State what could cause this error to occur.

.....

.....

Suggest a change to line 40 to overcome this problem.

.....

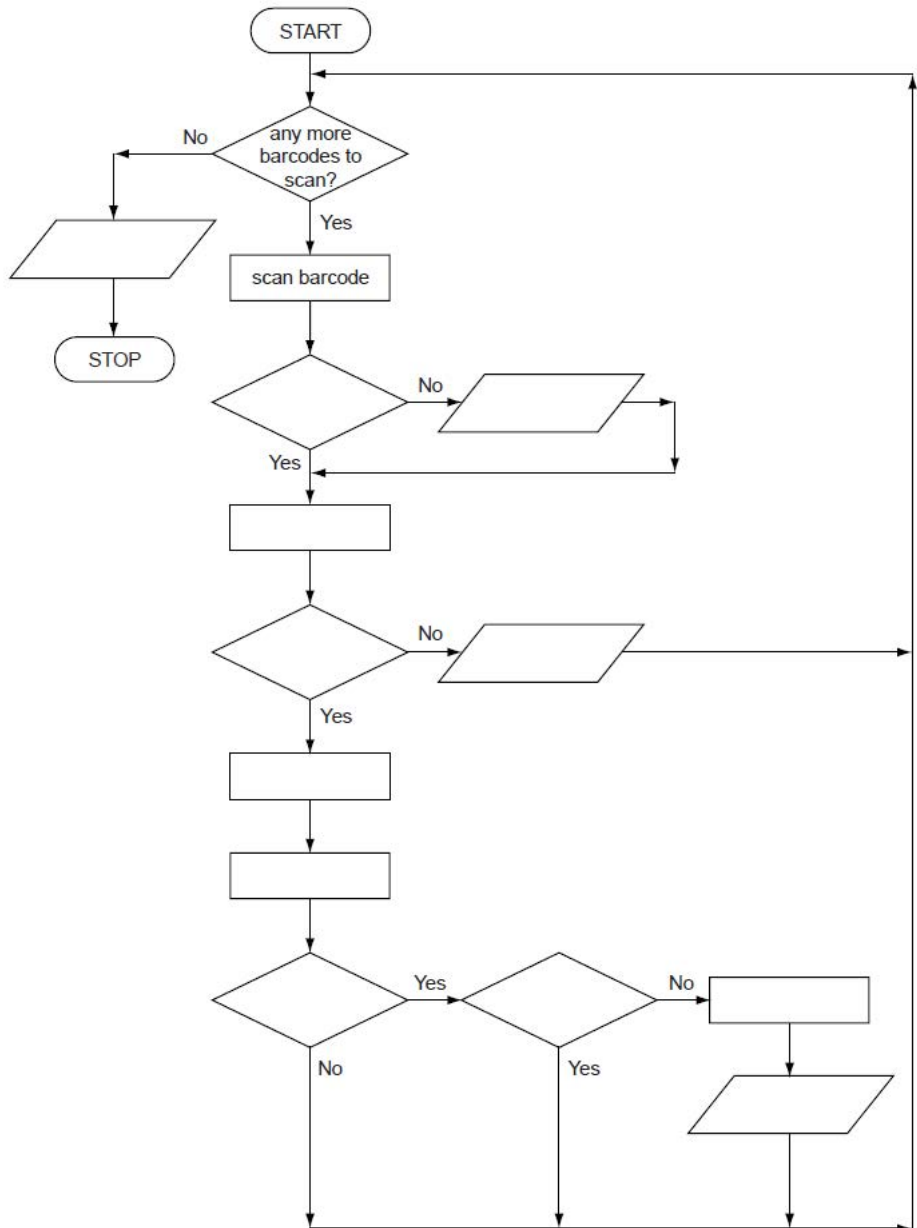
.....

- (a) The flowchart on the next page shows how barcodes are used in a supermarket to find product information and to produce orders for new stock automatically. Several statements are missing from the flowchart.

Question - 33

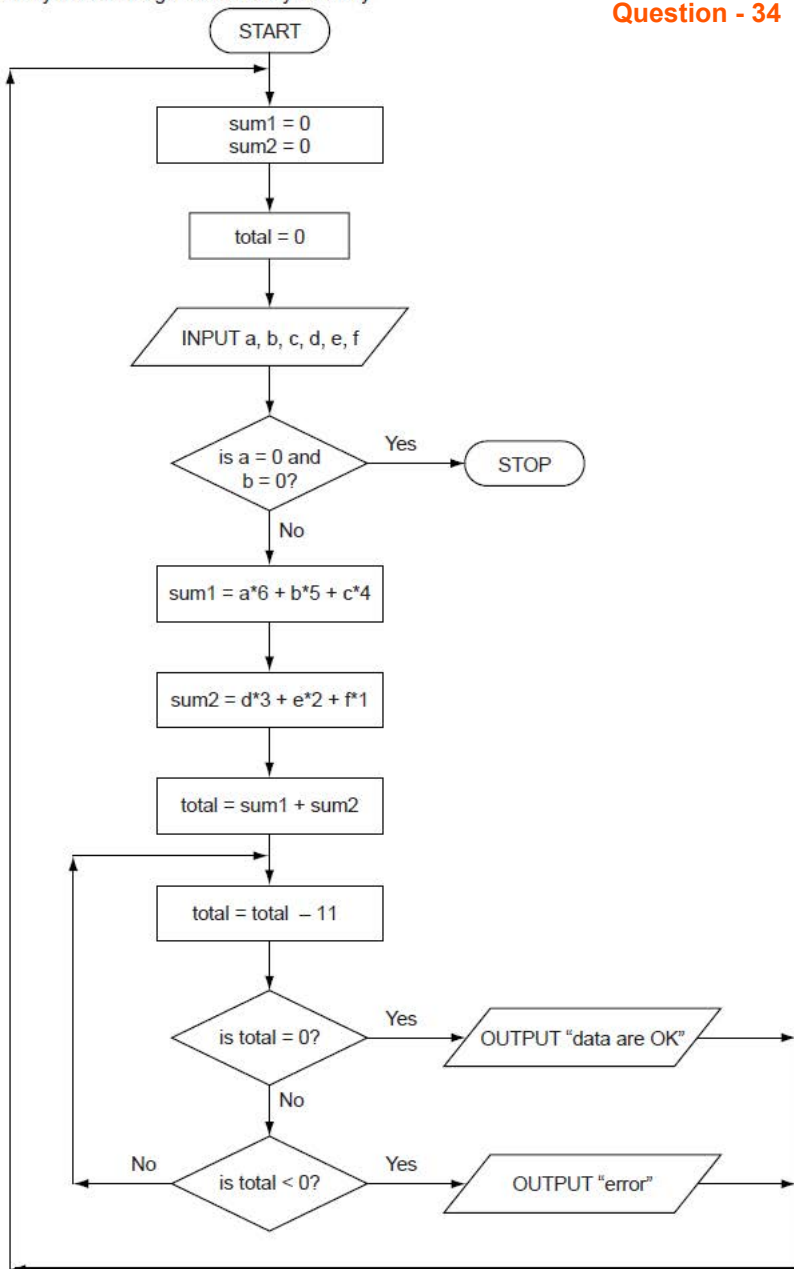
Complete the flowchart, using item numbers **only** from the list below.

Item number	Description
1	is barcode found?
2	is barcode read?
3	is flag for this product = 1?
4	is number in stock $\leq$ re-order value?
5	key in the barcode manually
6	locate price and product information from file
7	output an error message
8	output order request for new stock
9	output receipt and itemised bill
10	reduce number in stock by 1 and write new value back to the record
11	search database for barcode
12	set flag for this product to 1



Study the following flowchart very carefully.

Question - 34



A school has 3000 students sitting final examinations.
Each student sits eight examinations.

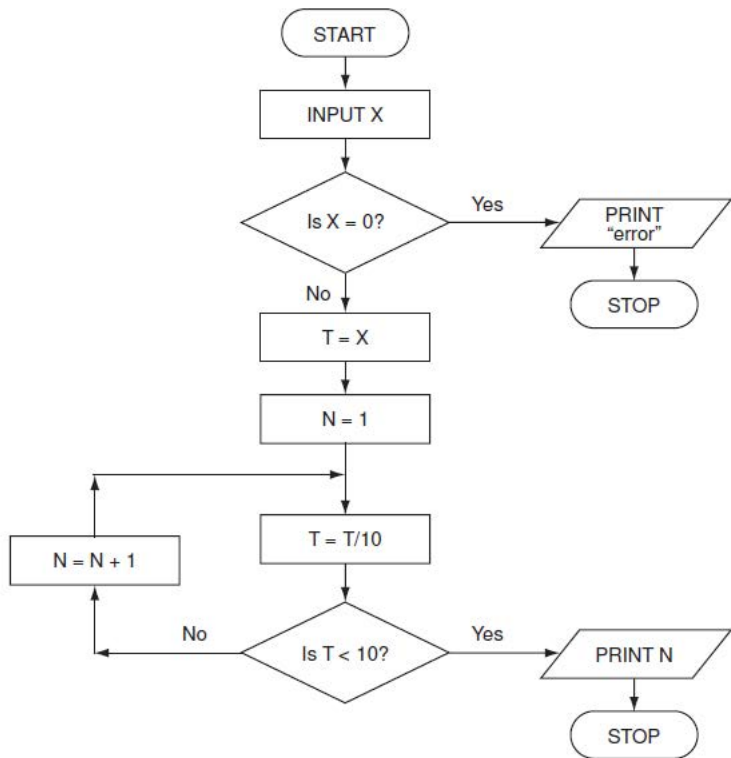
Question - 35

Write an algorithm, using pseudocode or a flowchart, which:

- inputs the marks for all 8 examinations for each student
- outputs for each student the average mark for their 8 examinations
- outputs the highest mark overall

Study the following flowchart very carefully:

Question - 36



What output would you expect if the following data was input into the flowchart?

X	OUTPUT
-150	
540	
0	

(a) Write an algorithm, using pseudocode or a flowchart, which:

Question - 37

- inputs 50 numbers
- outputs how many of the numbers were > 100

(a) Read the following section of code that inputs twenty (20) numbers and then outputs the largest number input.

Question - 38

```
1  h = 0
2  c = 0
3  repeat
4      read x
5      if x > h then x = h
6      c = c + 1
7      print h
8  until c < 20
```

There are **three** errors in this code.

Locate these errors and suggest a corrected piece of code.

1

2

3 [3]

(b) The above code is an example of a high-level language.

Give **two** features of a high-level language.

1

2 [2]

(c) The code is to be *interpreted* rather than *compiled*.

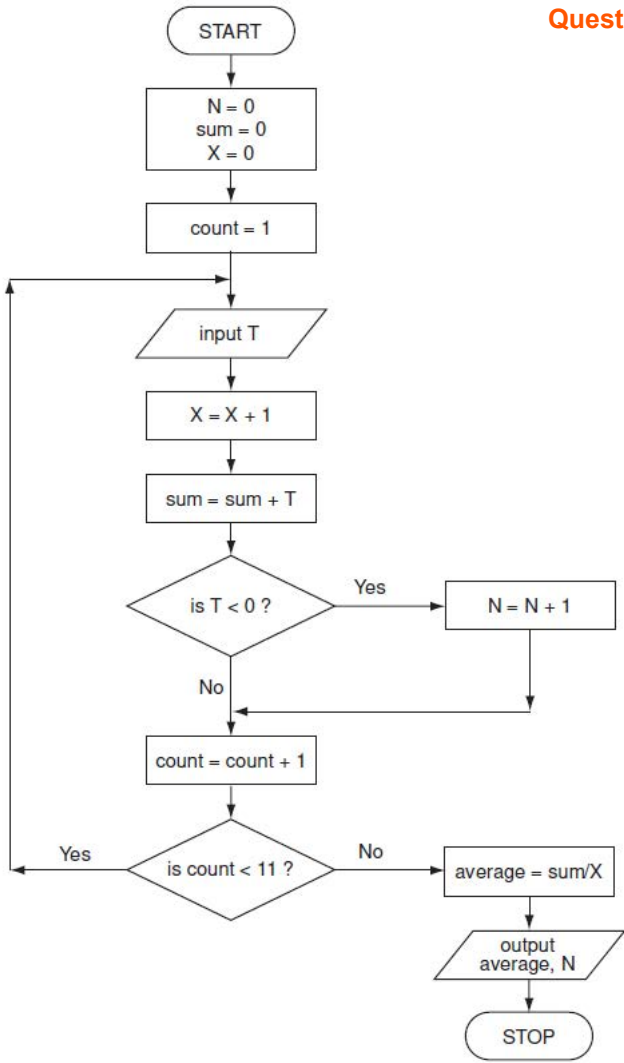
Give one difference between these two methods.

.....

.....

The following flowchart inputs ten temperatures and outputs the average (mean) temperature and the number of temperatures which were negative (i.e. < 0).

Question - 39



(a) Complete the trace table for this flowchart using the following test data:

5, 11, 16, -4, -10, 8, 10, -3, 17, 10

N	sum	X	count	T	average

[6]

(b) What values are output from the flowchart using the above test data?

.....

.....

[1]

A school has 1800 students. The start date and leaving date for each student is stored on file. Dates are in the format YYMMDD (e.g. a student starting on 10th September 2007 and leaving on 4th August 2012 has the data 070910 and 120804 on file).

(a) Write an algorithm, using pseudocode or otherwise, which

Question - 40

- inputs Student ID for all 1800 students
- inputs the start date and leaving date for each student
- carries out a check to ensure the second date is later
- if error, increments error counter
- outputs the number of errors

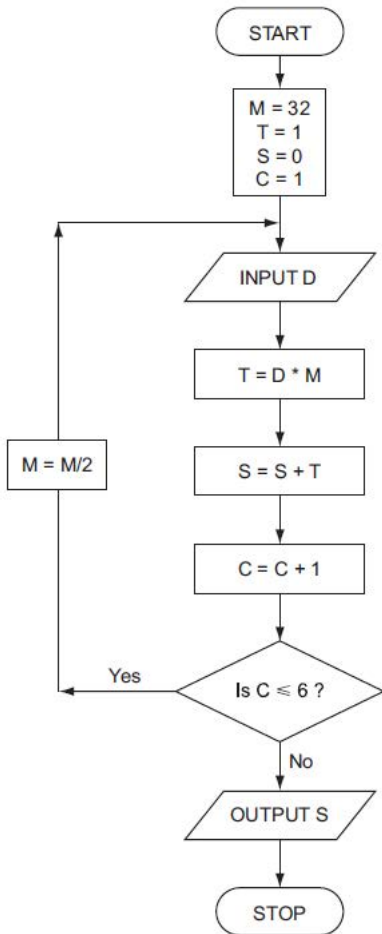
[5]

(b) Describe, with examples, **two** sets of test data you would use to test your algorithm.

[2]

Carefully study the following flowchart:

Question - 41



(a) Complete the trace table for the following data:

1, 0, 1, 1, 0, 1

M	T	S	C	D

[4]

(b) What process does this flowchart perform?

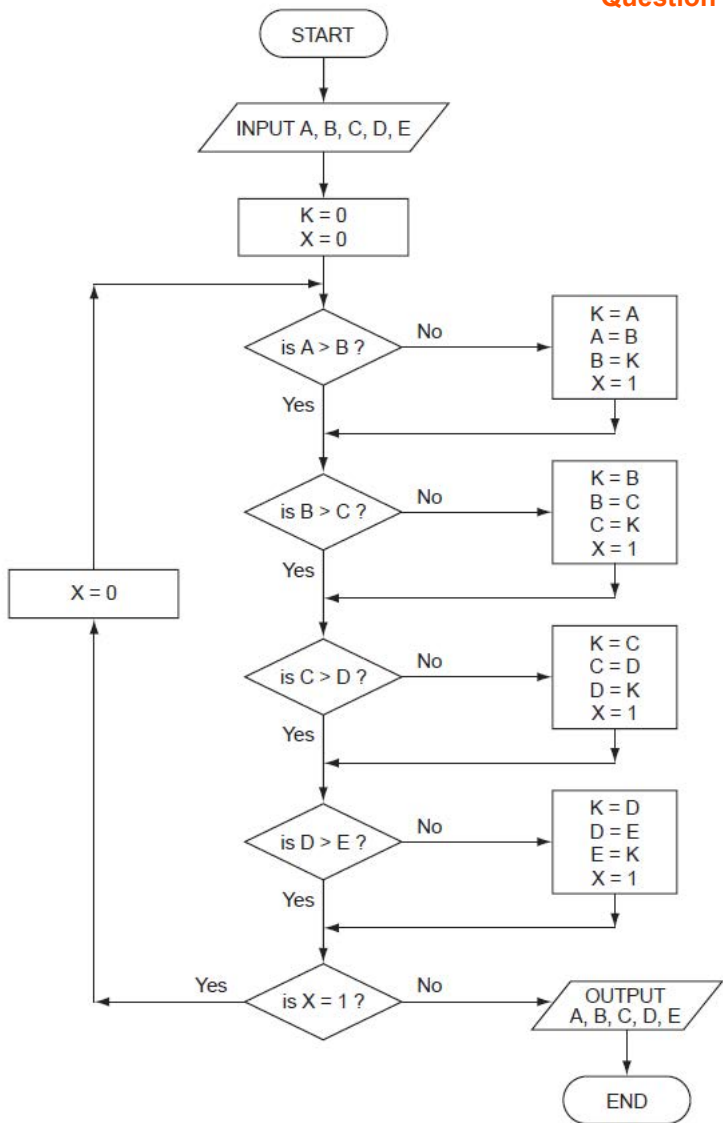
.....
.....
..... [1]

(c) Predict the output from the flowchart for an input of 1, 1, 1, 1, 0, 0

.....
..... [1]

Question - 42

- inputs the number of customer enquiries each day,
- inputs the house price each customer enquires about,
- outputs how many customers enquired each day about houses costing less than \$100 000,
- outputs the percentage of all enquiries made during the week about houses costing more than \$500 000.



(a) Complete the trace table for this flowchart using the following test data:

3, 5, 1, 4, 8

A	B	C	D	E	K	X

[5]

(b) What values are output from the flowchart using the above test data?

.....
..... [1]

(c) What function is this flowchart carrying out?

..... [1]

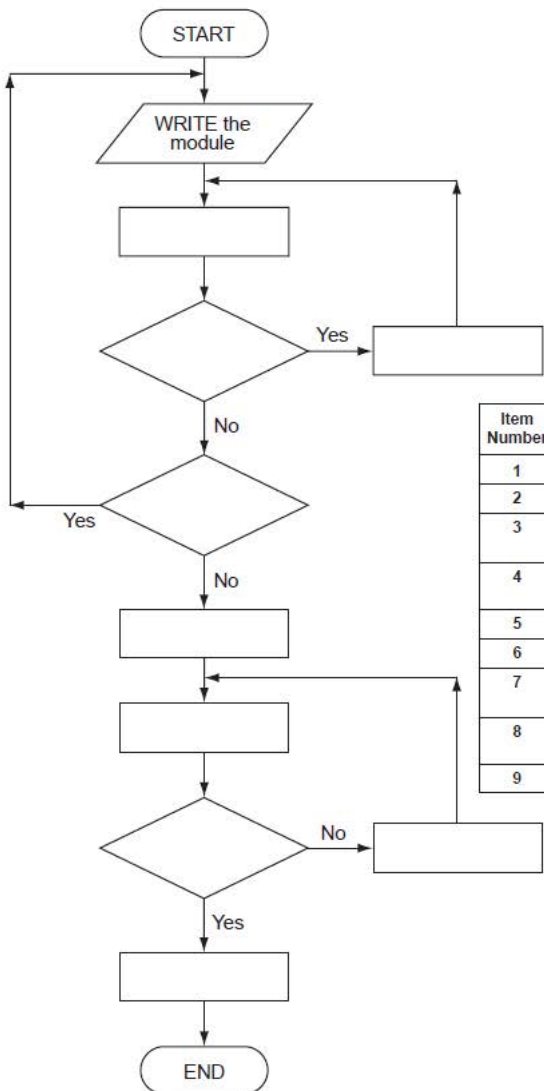
(d) What would happen if the value of X wasn't set to 0 in the return loop of the flowchart?

.....
..... [1]

16 A large word processor is being developed by first writing a series of modules. These are then put together to form the final word processor. Testing is done on each module and on the final word processor. The following flowchart shows how this word processor is developed. Several of the stages have been omitted.

Question - 44

Complete the flowchart, using item number only, from the list of items given.



ITEM LIST

Item Number	Item Description
1	any errors in the module?
2	any more modules to write and test?
3	combine all modules to form final word processor
4	does final word processor give expected results?
5	modify final word processor
6	modify the module
7	test module using data with known outcomes
8	test final word processor using data with known outcomes
9	write user documentation

The following algorithm should:

Question - 45

- input ten numbers
- output the largest number input
- output the average value of the input data

```
10    largest = 0
20    sum = 0
30    for x = 1 to 10
40        input x
50        if x > largest then x = largest
60        output largest
70        sum = sum + x
80    next x
90    average = sum * 10
100   output average
```

There are **four** errors in this algorithm.

Locate these errors and suggest a correction.

error 1

correction

error 2

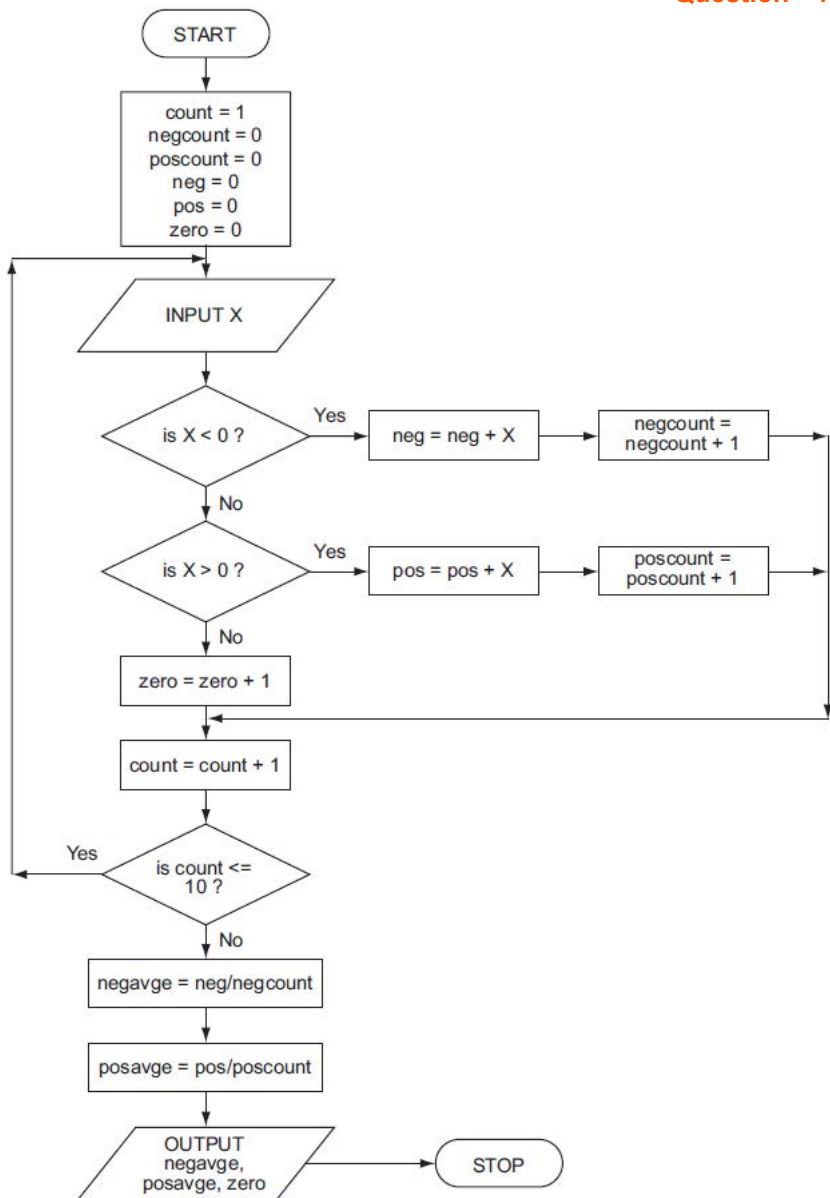
correction

error 3

correction

error 4

correction



Complete the trace table for the flowchart using the following data:

0, 3, 5, 6, -4, -1, 0, 0, -4, 10

negcount	poscount	neg	pos	zero	count	X	negavge	posavge

An algorithm has been written to check that code numbers are valid on input. They must be in the range 1000 to 9999.

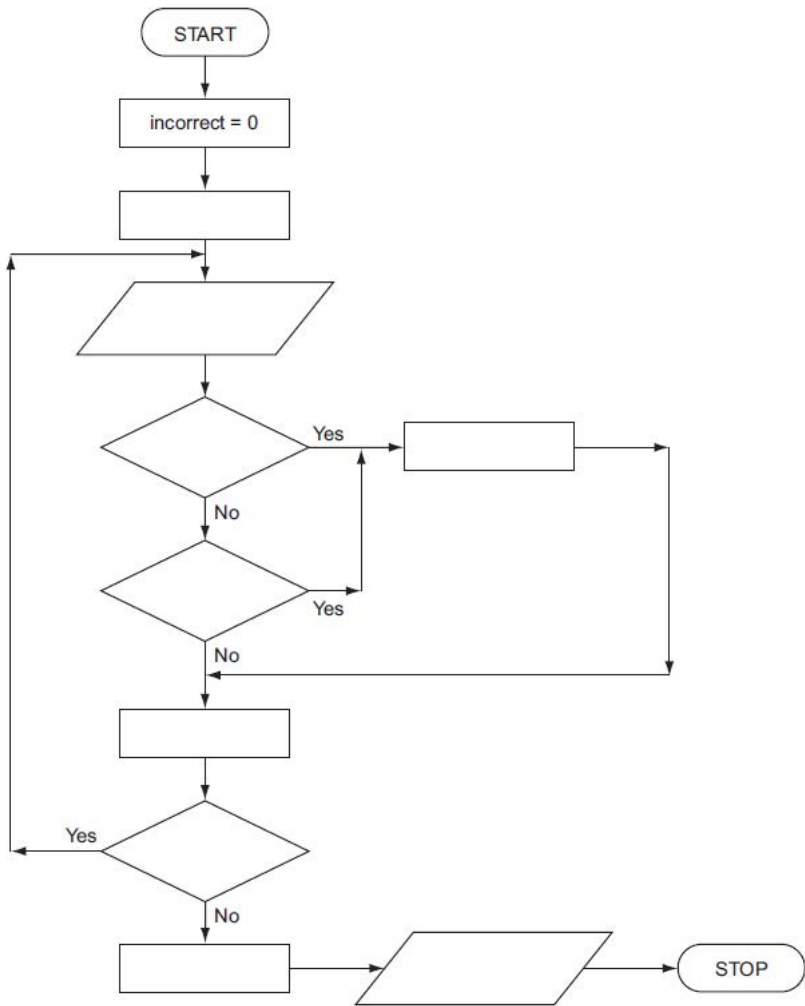
Question - 47

Five hundred codes are being entered and the percentage of entered codes which are incorrect is output.

There is a flowchart on the opposite page. It has some statements missing.

Complete the flowchart. Use statement numbers only, chosen from the list below.

statement number	statement
1	Incorrect = Incorrect + 1
2	INPUT Code
3	is Number ≤ 500 ?
4	is Code < 1000 ?
5	is Code > 9999 ?
6	Number = 1
7	Number = Number + 1
8	OUTPUT Percent
9	Percent = Incorrect / 5



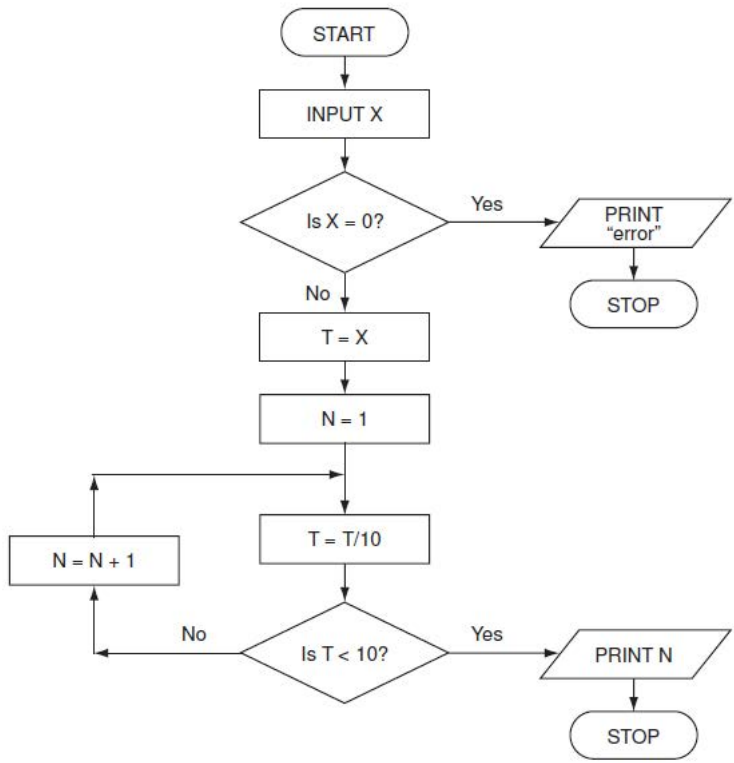
Question - 48

- Write an algorithm, using pseudocode or a flowchart, which:

- inputs the choice of all 1500 students (values 1 to 5)
- outputs all the names of the students who chose classical music
- outputs the percentage who chose **each** option.

Study the following flowchart very carefully:

Question - 49



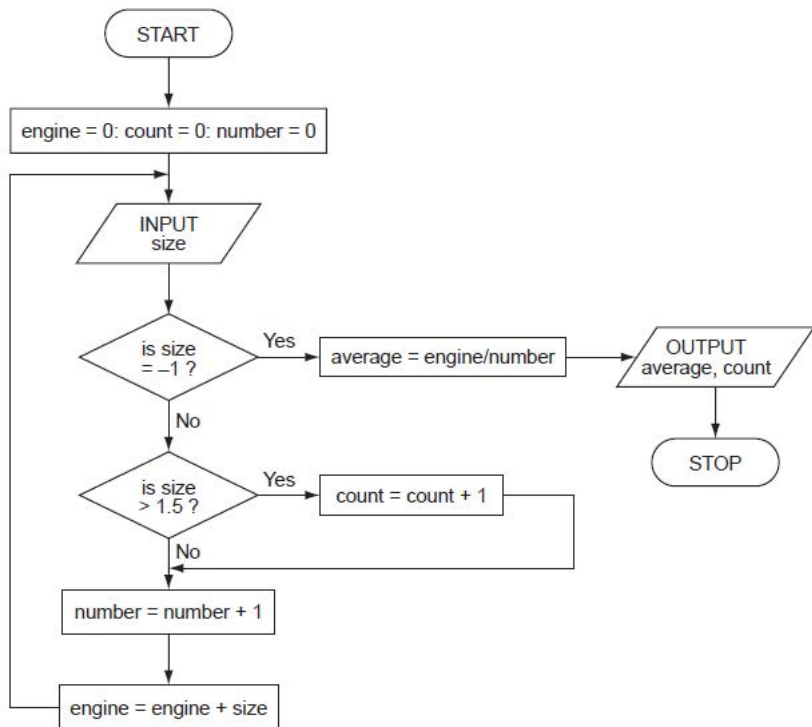
What output would you expect if the following data was input into the flowchart?

X	OUTPUT
-150	
540	
0	

The following flowchart inputs the size of a number of car engines; a value of -1 stops the input.

Question - 50

The following information is output: *average engine size* and *number of engines with size > 1.5*.



Complete the trace table for the following input data:

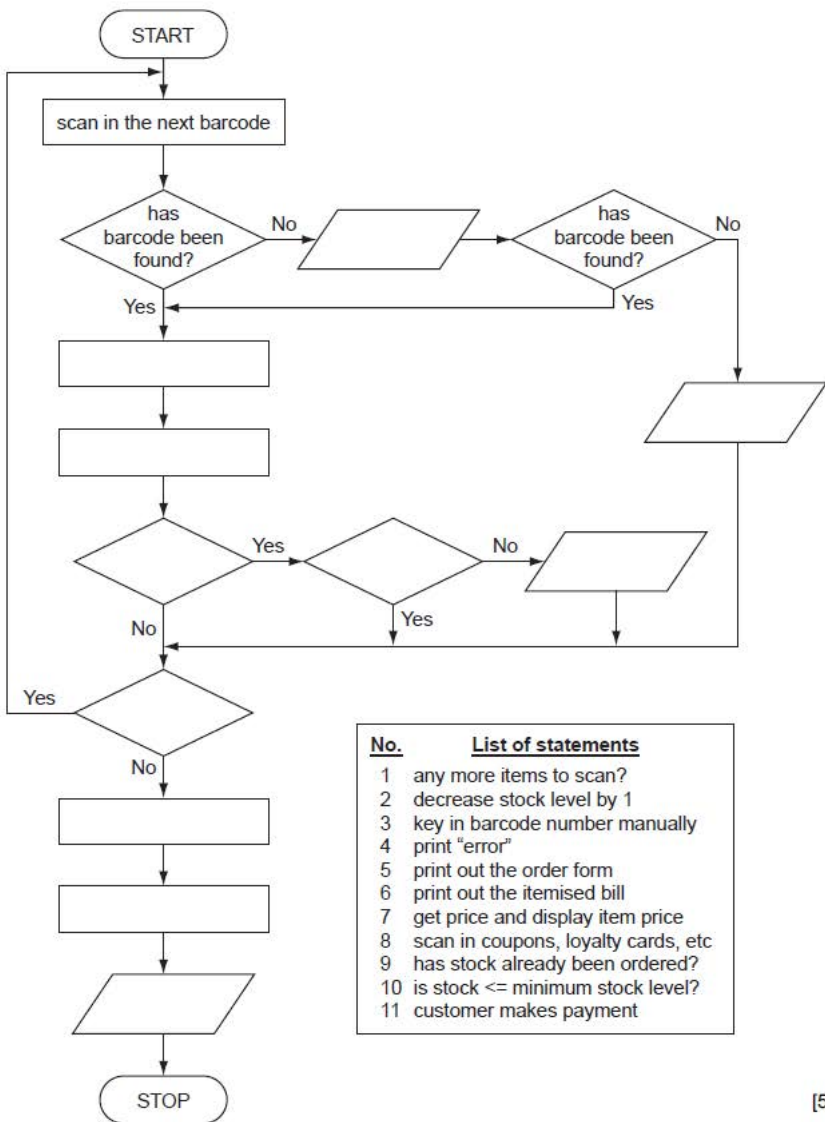
1.8, 2.0, 1.0, 1.3, 1.0, 2.5, 2.0, 1.3, 1.8, 1.3, -1

engine	count	number	size	average	OUTPUT

- 13 The following flowchart shows how barcodes are used when a customer buys items from the supermarket. Barcodes are used to find the prices and to carry out stock control. Several stages are missing from the flowchart. These stages are shown in the item list below.

Question - 51

Insert the correct statements, using numbers only, to complete the flowchart.

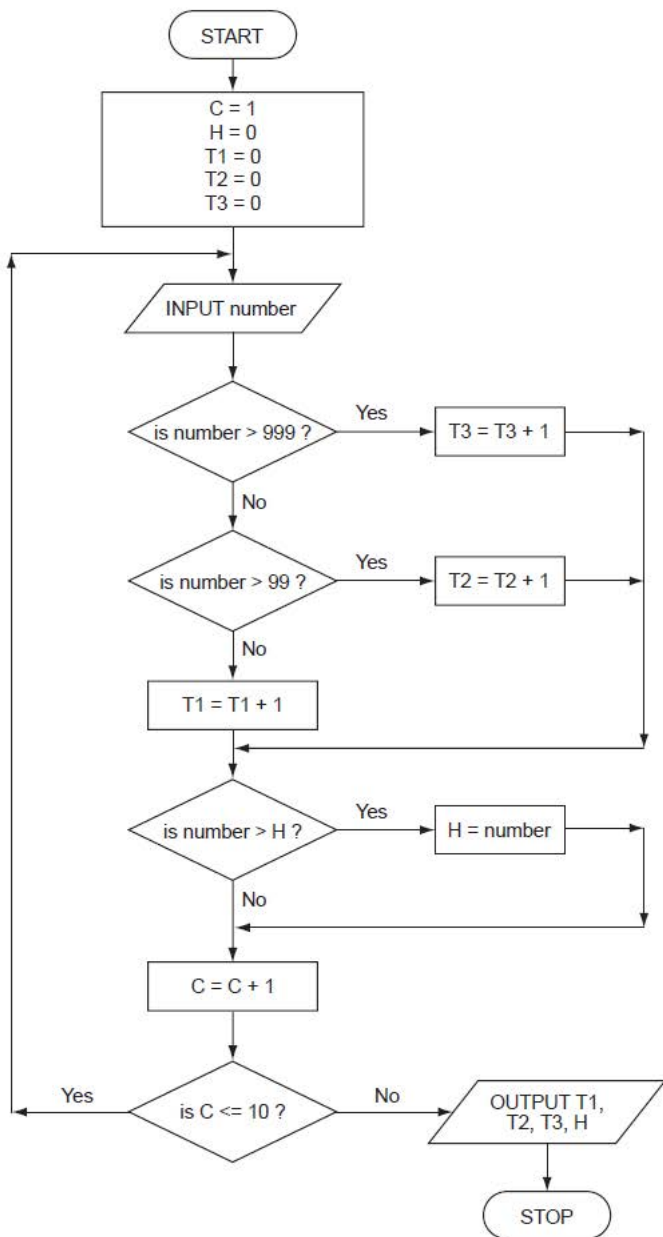


The weather conditions in a town are being monitored over a year (365 days). The values recorded **per day** are weather type and temperature (e.g. CLOUDY, 25)

Question - 52

Write an algorithm, using pseudocode or flowchart only, which:

- inputs the weather type and temperature for **each day**
- outputs the number of days that were CLOUDY, RAINING, SUNNY or FOGGY
- outputs the highest recorded temperature for the year
- outputs the lowest recorded temperature for the year



Complete the trace table for the following data:

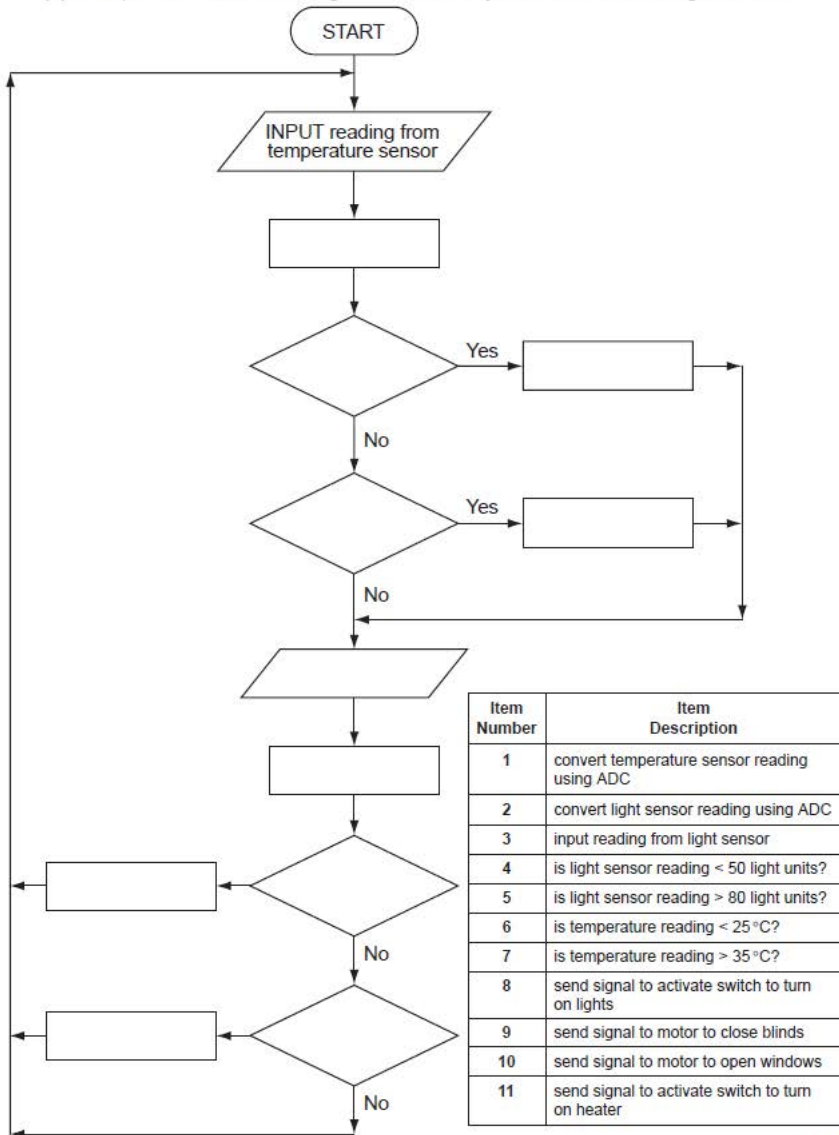
1500, 1000, 100, 10, 999, 99, 2000, 5, -3, 0

C	H	T1	T2	T3	number	OUTPUT

- 14 The following flowchart shows how a computer and sensors are used to control the environment in a greenhouse. Temperatures must be between 25°C and 35°C. Light must be between 50 and 80 light units.

Question - 54

- (a) Complete the flowchart, using item number only, from the list of items given below.

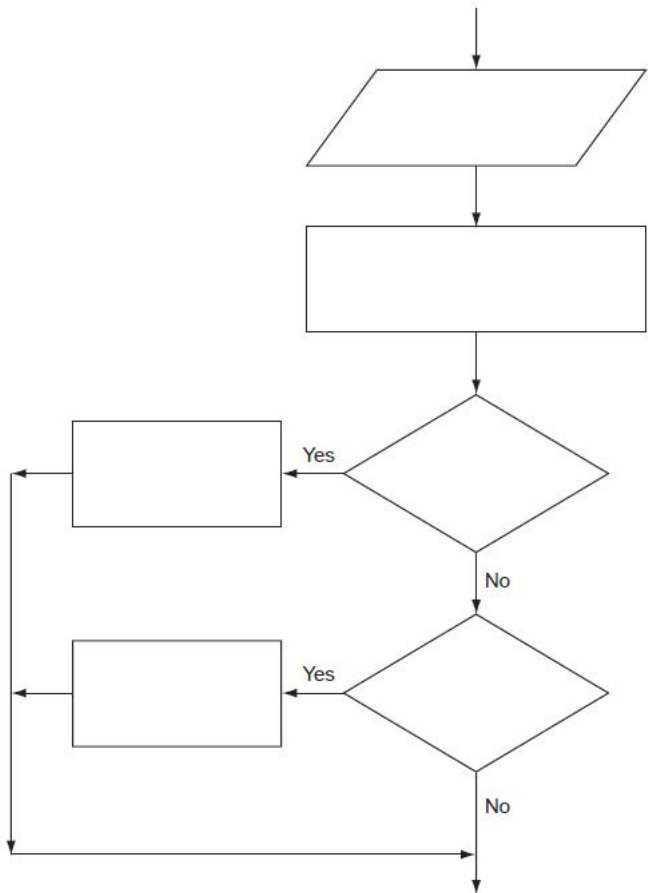


- (b) The computer also checks on humidity levels (using humidity sensors) which must be between the values of 40 and 90.

If humidity is too low, water is sprayed into the air.

If humidity is too high, fresh air is allowed to enter.

Write the necessary commands in the following flowchart section to show how the humidity levels are controlled:



A small café sells five types of item:

bun	0.50	dollars
coffee	1.20	dollars
cake	1.50	dollars
sandwich	2.10	dollars
dessert	4.00	dollars

Question - 55

Write an algorithm, using pseudocode or a program flowchart only, which

- inputs every item sold during the day,
- uses an item called "end" to finish the day's input,
- adds up the daily amount taken for each type of item,
- outputs the total takings (for all items added together) at the end of the day,
- outputs the type of item that had the highest takings at the end of the day.

(a) The following pseudocode was written to input 1000 dates.

```

1  count = 1
2  repeat
3      input day, month, year
4      count = count + 1
5  until count = 1000

```

(i) Describe why the loop only inputs 999 dates instead of 1000.

[1]

(ii) What needs to be changed or added to the above code to make sure 1000 dates are input?

[1]

(b) Errors in code can be found using test data.

Name **three** different types of test data. Using `month` from the pseudocode above, give an example of each type of test data.

test data type 1

example

test data type 2

example

test data type 3

example

[6]

A microprocessor controls the opening and closing of automatic doors to a supermarket.

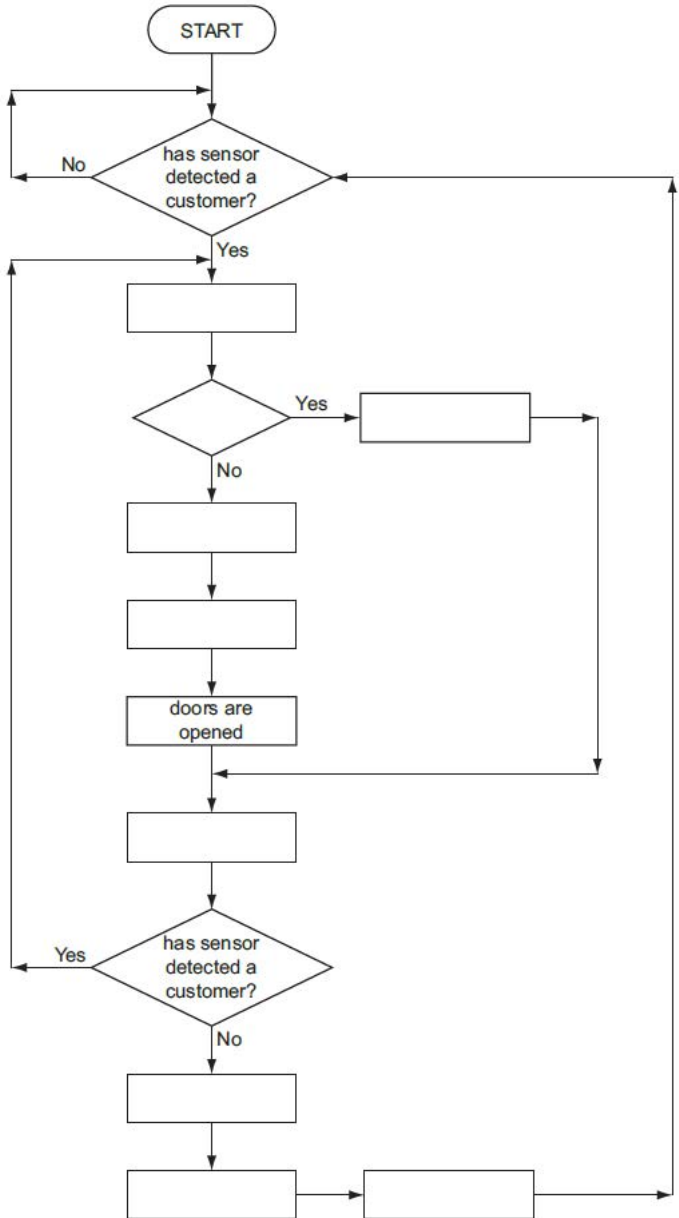
Customers are detected using pressure sensors.

Question - 57

The flowchart on the next page shows how the sensors and microprocessor interact to control the opening and closing of the doors. However, several of the stages in the process have been missed out.

Using **item number only**, complete the flowchart using items from the following list:

Item number	Item description
1	ADC changes analogue signal into digital signal
2	are the doors already open?
3	DAC changes digital signal into analogue signal
4	DAC changes digital signal into analogue signal
5	doors are closed
6	microprocessor sends signal
7	microprocessor sends signal
8	no action is required
9	wait for 10 seconds



5000 numbers are being input which should have either 1 digit (e.g. 5), 2 digits (e.g. 36), 3 digits (e.g. 149) or 4 digits (e.g. 8567).

Question - 58

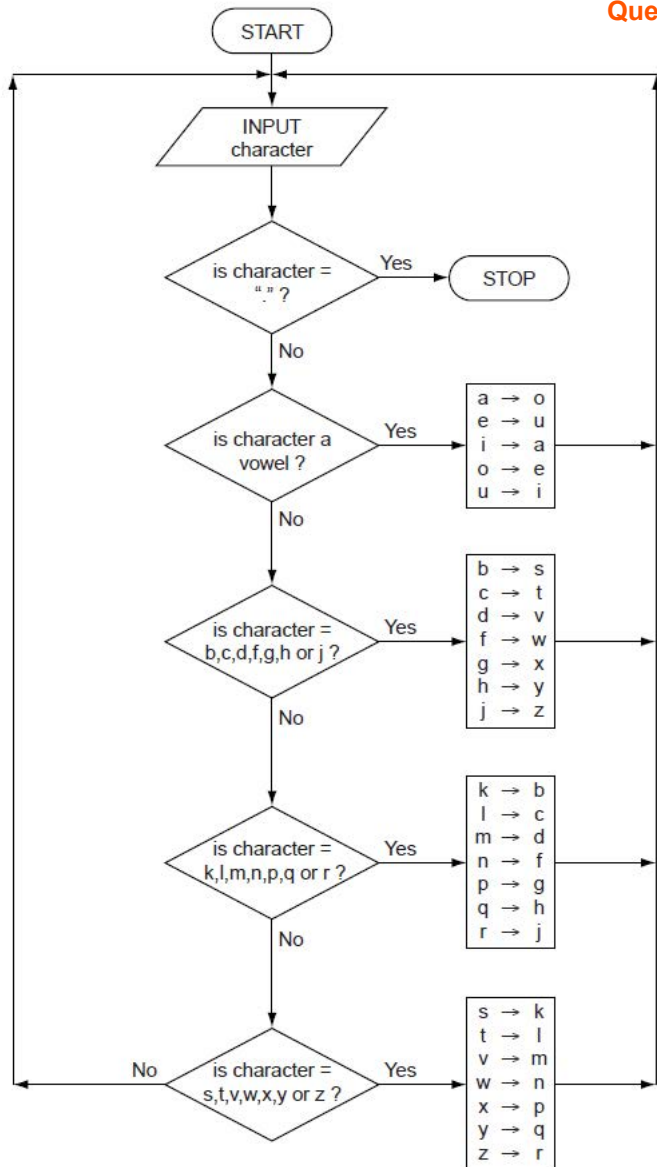
Write an algorithm, using pseudocode or flowchart only, which

- inputs 5000 numbers
- outputs how many numbers had 1 digit, 2 digits, 3 digits and 4 digits
- outputs the % of numbers input which were outside the range

[illegible]

Data sent across the Internet are frequently encrypted. The following flowchart shows a basic encryption method (Note: the → symbol in the flowchart means “is replaced by”).

Question - 59



For example,

h e l l o i a m a r o b o t .

becomes:

y u c c e a o d o j e s e l .

(a) Use the flowchart to encrypt the following message:

m e e t i n g w i l l g o a h e a d .

[2]

(b) Use the information in the flowchart to show which input message produced the following encrypted message:

t e d g i l a f x a k w i f .

[2]

(c) Many customers shop online.

Apart from encryption, describe **three** other security features built into many online shopping websites.

1

2

3

[3]

An algorithm has been written to input six temperatures for every day of the year (365 days). The outputs are:

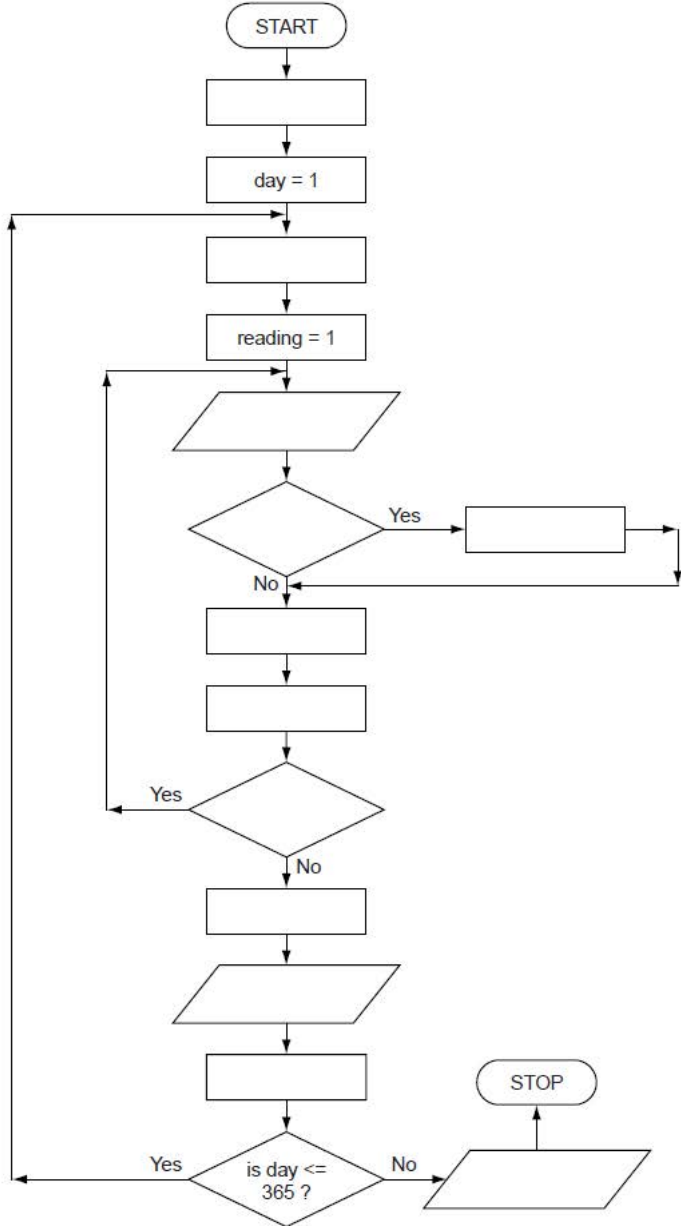
Question - 60

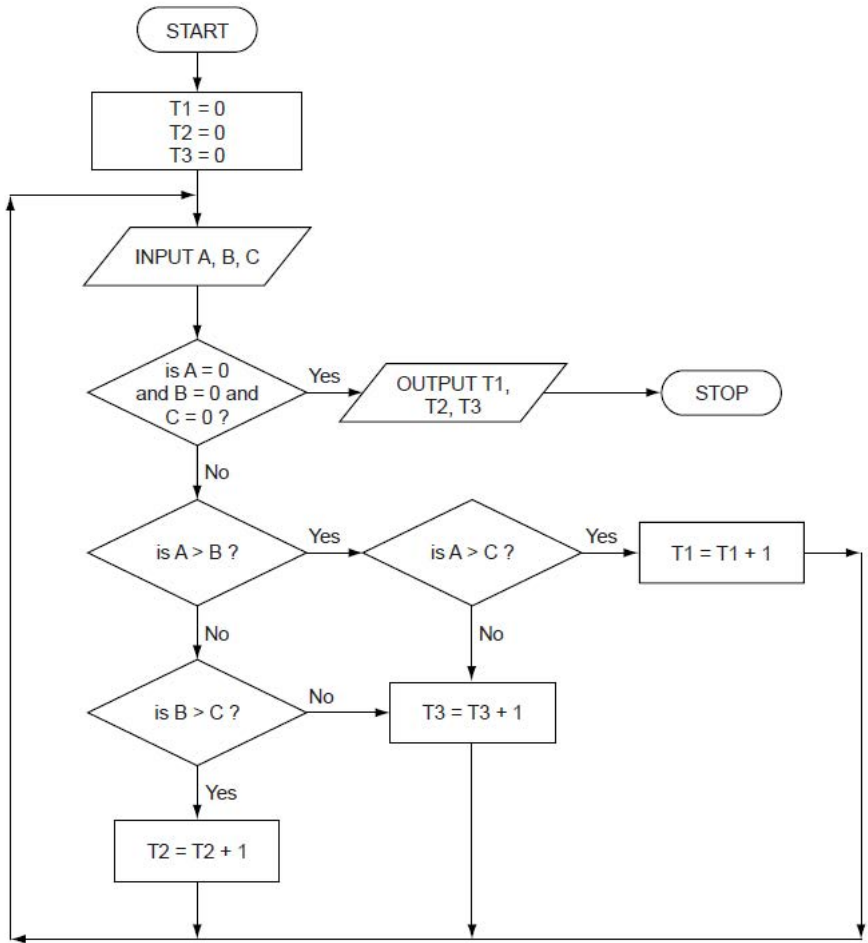
- the average daily temperature for each day
- the highest recorded temperature for the whole year

The algorithm is in the form of a flowchart on the next page. However, several of the statements are missing.

Using instruction number **only**, complete the flowchart using the following list of instructions:

Instruction number	Instruction
1	average = total / 6
2	day = day + 1
3	high = -200
4	high = temperature
5	input temperature
6	is reading <= 6 ?
7	is temperature > high ?
8	output average
9	output high
10	reading = reading + 1
11	total = 0
12	total = total + temperature





(a) Complete the trace table for the flowchart using the following data:

3, 2, 1

4, 8, 7

6, 0, 3

5, 6, 9

4, 11, 3

0, 0, 0

T1	T2	T3	A	B	C	OUTPUT

[5]

(b) This flowchart does not give correct answers for certain sets of test data.

Suggest a data set that would give an incorrect answer.

Give a reason for your choice.

data set

.....

reason

.....

.....[2]

A customer wants to compare prices of 1000 items sold in two supermarkets (price1 and price2).

Write an algorithm, using pseudocode or a flowchart, which:

Question - 62

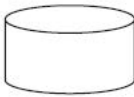
- inputs the two prices for all 1000 items
- outputs how many items were more expensive in supermarket 1
- outputs how many items were more expensive in supermarket 2
- outputs the largest price difference

Identify what you would use the following system flowchart symbols for.

Question - 63



A



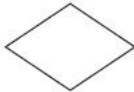
B



C



D



E



F



G



H

A
B
C
D
E
F
G
H